BUSINESS ANALYTICS PROJECT

I built an ML model that predicts customer churn with 87% AUC

Here's what I found —
and how I built it in Python

Swipe →

The Business Problem

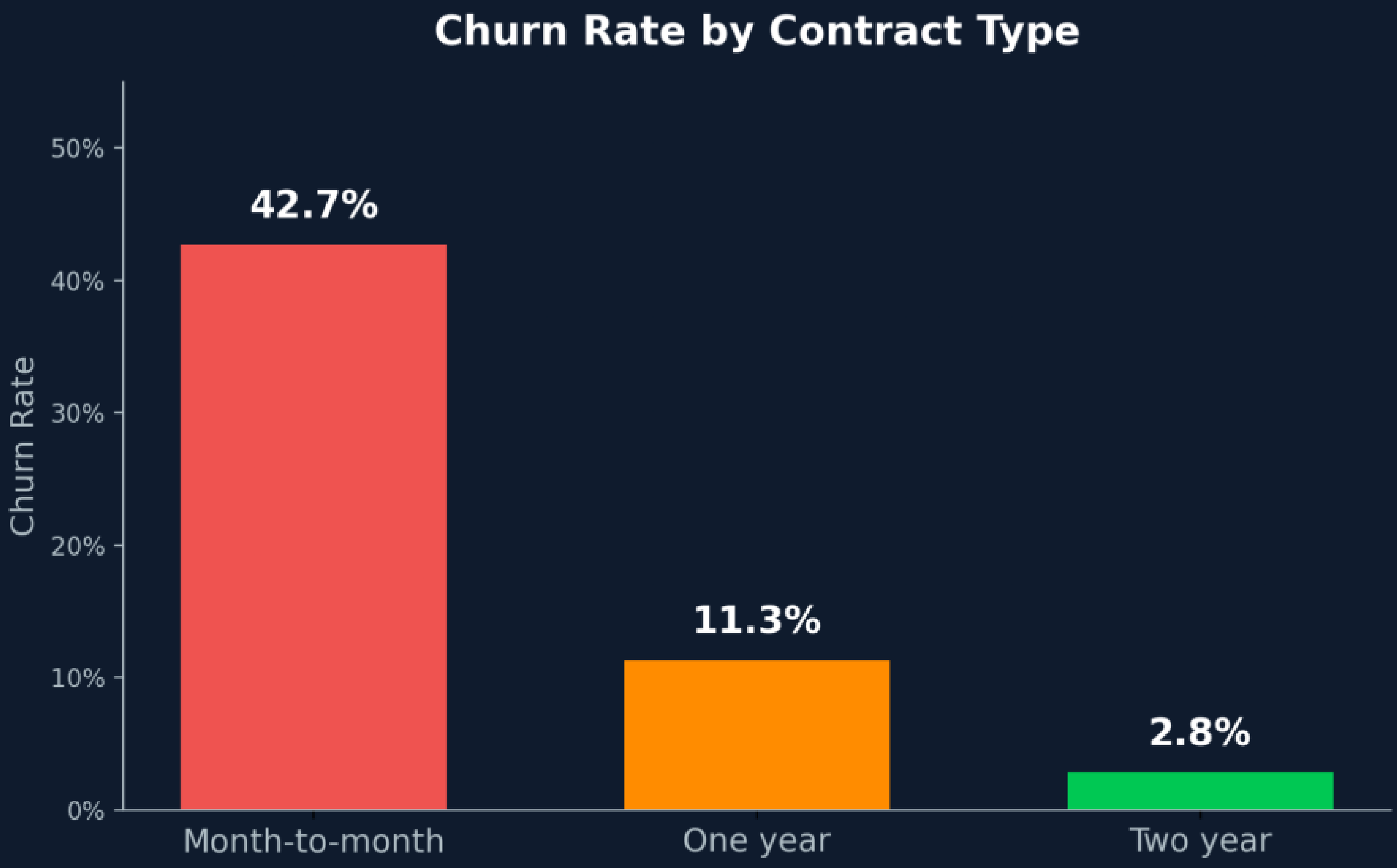
26.5% of customers churned

The goal: predict who leaves before they do,
so the retention team can intervene.

Dataset

IBM Telco • 7,043 customers • 20 features

The Biggest Churn Driver? Contract Type.



Locking customers into longer contracts is the single biggest retention lever.

Model Comparison

Model	AUC	
Logistic Regression	0.83	
Random Forest	0.85	
XGBoost	0.87	← Best

XGBoost correctly ranks a churner above a non-churner 87% of the time.

What This Means in Practice

At 80% recall, the model flags ~8 in 10 churners before they leave.

Recoverable Revenue

£137,000 / year

Based on £65/month avg revenue • 1,000 customers

Full code + app on GitHub
link in comments ↓